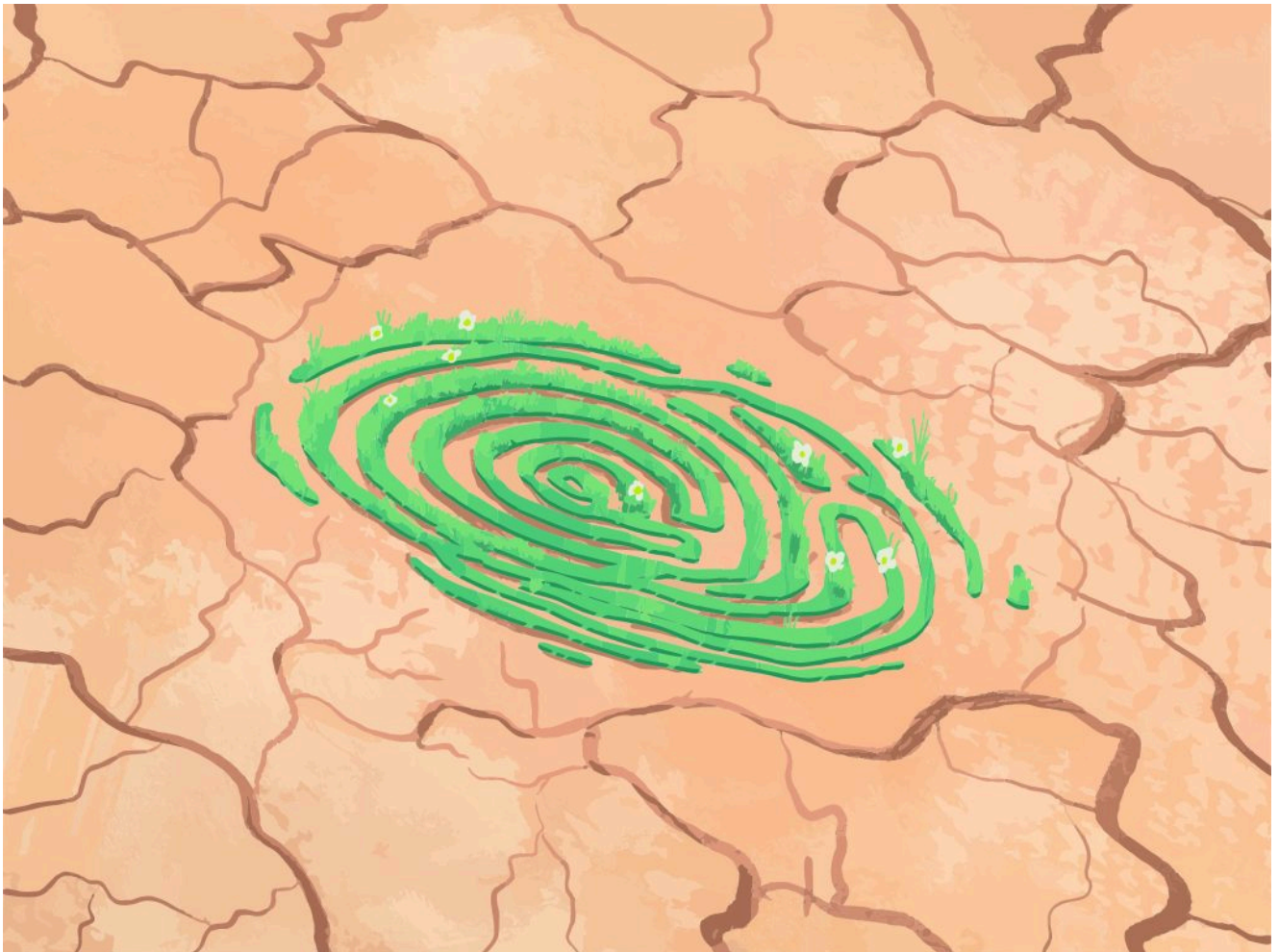


KUBECON CLOUDNATIVECON EU 2026 / KUBERNETES / СЕТЕВЫЕ ТЕХНОЛОГИИ / БЕЗОПАСНОСТЬ

Почему плоские сети Kubernetes не масштабируются

Узнайте, как внедрение иерархии безопасности и имитации политик в сетях Kubernetes повышает качество подключения, безопасность и соответствие требованиям.

20 марта 2026 года, 7:00 [Реза Рамезанпур](#)



Ардиан Праномо для Unsplash+

[CNCF](#) спонсировала этот пост.

Переосмысление иерархии сетевой безопасности для облачных платформ
Сетевые возможности Kubernetes впечатляют. Гибкость этой платформы воляет командам соединять сотни микросервисов в разных пространствах имен, кластерах и средах. Но по мере роста платформ эта же

Для многих организаций проблемы в первую очередь возникают при работе с сетями. Инженеры с трудом устраняют неполадки с подключением. Команды по обеспечению безопасности сталкиваются с трудностями при внедрении глобальных средств контроля. Архитекторы платформ вынуждены доказывать соответствие требованиям. И большинство этих проблем имеют одну и ту же первопричину: **плоские модели сетевой безопасности, которые не масштабируются.**

Ограничения плоских сетей

Kubernetes NetworkPolicy дает командам возможность контролировать трафик между рабочими нагрузками. По умолчанию все политики существуют на одном уровне без встроенного управляемого приоритета.

«По мере развития политики становится все труднее предсказать, что произойдет, если вы что-то измените».

Это хорошо работает в небольшом кластере, состоящем из одной команды. Но в крупных средах с несколькими командами такой подход быстро становится рискованным.

В плоской модели безопасность обеспечивается за счет исключений, а не принудительных мер. Защита критически важного сервиса часто сводится к перечислению всех разрешенных подключений и надежде, что ничто другое случайно не нарушит это правило. По мере усложнения политик становится все труднее предсказать, что произойдет при внесении изменений.



Cloud Native Computing Foundation (CNCF) размещает критически важные компоненты глобальной технологической инфраструктуры, в том числе Kubernetes, OpenTelemetry и Argo. CNCF — это нейтральная площадка для совместной работы в сфере облачных технологий,

THE LATEST FROM CNCF

Building a Cluster-Aware AI Agent with Kubernetes, Argo CD, and GitOps

25 June 2026

From Awareness to Engineered Accessibility in Open Source

24 June 2026

Building Jaeger's ClickHouse backend: 8.6× compression on 10 million spans

23 June 2026

Without clear rules for precedence or tools for validation, troubleshooting becomes a detective exercise. Teams constantly ask: *Which policy ran first? Which rule actually applied? Did a recent change break a security control?*

Change gridlock and compliance pressure

These issues directly affect day-to-day operations.

TRENDING STORIES

1. **A year in, Google wants its Axion processors to feel like a scheduling decision**
2. **What is KubeVirt and why it's growing**
3. **Amazon EKS Auto Mode wants to end Kubernetes toil — one node at a time**
4. **Sampling: the philosopher's stone of distributed tracing**
5. **Why flat Kubernetes networks fail at scale**

When teams can't confidently answer "What happens if I apply this policy?", change feels risky. Policies get delayed, avoided, or applied conservatively. Over time, this leads to policy drift, technical debt, and bigger attack surfaces.

show, leading to ad-hoc headaches and sometimes extra work outside the cluster.

The result? Change gridlock. Networking becomes a bottleneck instead of a foundation for innovation, slowing delivery and adding stress for everyone.

Bringing structure with security hierarchies

The solution is simple in principle: introduce hierarchy.

A security hierarchy gives network policies explicit order and separation of responsibility. Instead of all rules competing at the same level, policies are grouped and evaluated by priority and purpose.

Common patterns include:

- **Platform tiers** – required connectivity for cluster services
- **Security tiers** – mandatory controls such as egress restrictions or deny rules
- **Application tiers** – developer-managed rules for service communication
- **Data or infrastructure tiers** – protecting high-value workloads such as databases

Security team	Tier	Platform team	Tier	Dev team	Tier
quarantine		logging		dmz	
allow-tigera-restricted resources		twilio integration		trusted	
block-feodo		allow kube-dns		restricted	
pki-whitelist		staged-test		quarantine	

A practical implementation of a security hierarchy, showing how different teams manage specific tiers and policies to maintain a Zero Trust posture.

 Zoom

Hierarchies make policy intent clear and reduce accidental overrides. Global rules are enforced consistently, while teams retain autonomy within defined boundaries. This approach also aligns with **Zero Trust principles**, where access is explicitly granted and continuously evaluated, even inside the cluster.

sting changes without breaking things

Hierarchy alone is not enough. Teams also need safe ways to test new policies.

A growing best practice is policy simulation or dry-run mode. Policies are deployed without enforcing them. The system observes traffic and reports what would have been allowed or denied.

This allows teams to:

- Safely validate new rules against live workloads
- Refine policies based on real data
- Collaborate earlier across platform, security, and application teams

By moving validation earlier in the lifecycle, organizations reduce outages and speed up secure change.

A broader trend in cloud-native security

Moving away from flat networks reflects a wider shift in the cloud-native community.

As **platforms get bigger and more complex**, teams are looking for:

- Clear separation of responsibilities
- Predictable behavior across environments
- Tooling that supports intent-driven configuration instead of manual coordination

Hierarchies and dry-run testing are becoming standard patterns. They are not tied to a single tool and are useful across multiple environments. They are especially important as organizations adopt AI workloads, hybrid deployments, and clusters at global scale.

Where Kubernetes networking is headed

Flat network models were fine when clusters were small and teams were tightly coupled. That is no longer the case.

behavior predictable, auditable, and resilient.”

To operate Kubernetes securely at scale, platform teams are reintroducing structure through hierarchies and safe change mechanisms. The goal is not to slow innovation. The goal is to make network behavior predictable, auditable, and resilient.

As the cloud-native ecosystem continues to evolve, these patterns will turn networking from a source of risk into a reliable foundation for modern platforms.

*This guest column is being published ahead of **KubeCon + CloudNativeCon Europe**, the Cloud Native Computing Foundation’s flagship conference, which will bring together adopters and technologists from leading open-source and cloud-native communities in Amsterdam, the Netherlands, from March 23-26, 2026.*

TNS



Reza Ramezani is a Senior Developer Advocate at Tigera, working to promote the widespread adoption of Kubernetes and Project Calico. With over 15 years of experience in the IT industry, Reza’s expertise ranges from reverse engineering to full stack development....

[Read more from Reza Ramezani →](#)

SHARE THIS STORY



TRENDING STORIES

1. A year in, Google wants its Axion processors to feel like a scheduling decision
 2. What is KubeVirt and why it’s growing
 3. Amazon EKS Auto Mode wants to end Kubernetes toil — one node at a time
- Sampling: the philosopher's stone of distributed tracing

INSIGHTS FROM OUR SPONSOR



The Cloud Native Computing Foundation (CNCF) hosts critical components of the global technology infrastructure including Kubernetes, OpenTelemetry, and Argo. CNCF is the neutral home for cloud native collaboration, bringing together the industry's top developers, end users, and vendors.

[Learn More →](#)

Building a Cluster-Aware AI Agent with Kubernetes, Argo CD, and GitOps

25 June 2026

From Awareness to Engineered Accessibility in Open Source

24 June 2026

Building Jaeger's ClickHouse backend: 8.6× compression on 10 million spans

23 June 2026

Agent Auth: A lawyer's day in court

23 June 2026

Telemetry that matters: Designing sustainable, high-impact observability pipelines

22 June 2026

KubeCon + CloudNativeCon, OpenInfra Summit and PyTorch Conference Unite in China to Scale AI

18 June 2026

most recent TNS articles in your inbox each day.

svo1975pvn1982@gmail.com

SUBSCRIBE

The New Stack does not sell your information or share it with unaffiliated third parties. By continuing, you agree to our [Terms of Use](#) and [Privacy Policy](#).

ARCHITECTURE

Cloud Native Ecosystem
Containers
Databases
Edge Computing
Infrastructure as Code
Linux
Microservices
Open Source
Networking
Storage

ENGINEERING

AI
AI Engineering
API Management
Backend development
Data
Frontend Development
Large Language Models
Security
Software Development
WebAssembly

OPERATIONS

AI Operations
CI/CD
Cloud Services
DevOps
Kubernetes

CHANNELS

Podcasts
Ebooks
Events
Webinars
Newsletter

THE NEW STACK

[About / Contact](#)
[Sponsors](#)
[Advertise With Us](#)
[Contributions](#)



Community created roadmaps, articles, resources and journeys for developers to help you choose your path and grow in your career.

[Frontend Developer Roadmap](#)
[Backend Developer Roadmap](#)
[Devops Roadmap](#)

FOLLOW TNS



© The New Stack 2026

[Disclosures](#)
[Terms of Use](#)
[Advertising Terms & Conditions](#)
[Privacy Policy](#)
[Cookie Policy](#)